INCH-POUND

MIL-A-53100(ME)
7 February 1990

MILITARY SPECIFICATION

ARCTIC FUEL SYSTEM SUPPLY POINT (AFSSP)

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an arctic fuel system which consists of two modules. Each module consists of three collapsible fabric tanks (in any combination of 3,000, 10,000 and 20,000 gallon capacities); two 600 GPM (gallons per minute) pumps; and three 200 GPM filter-separators; and ancillary equipment. A module is capable of receiving, dispensing, and storing automotive gasoline, aviation gasoline, jet fuel, or diesel fuel at all ambient temperatures from -60 to +95 °F.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1108	- Extinguisher, Fire, Vaporizing Liquid.
QQ-S-766	- Steel, Plates, Sheets and Strip - Corrosion Resisting.
QQ-S-781	- Strapping, Steel, and Seals.
VV-F-800	- Fuel Oil, Diesel.
WW-C-440	- Clamps, Hose (Low Pressure).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-V-3	- Valves, Fittings, and Flanges (Except for Systems Indicated Herein): Packaging of.
MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-H-775	- Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing) and Associated Hardware: Packaging of.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-C-5501	- Cap and Plug, Protective, Dust and Moisture Seal.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-C-27487	- Coupling Halves, Quick-Disconnect, Cam-Locking Type.
MIL-C-46168	- Coating, Aliphatic, Polyurethane, Chemical Agent Resistant.
MIL-F-53091	- Filter-Separator, Liquid Fuel: Frame Mounted, 200 GPM Capacity, Arctic Service.
MIL-T-53101	- Tanks, Fabric, Collapsible: 3,000, 10,000 and 20,000 Gallon Petroleum, Low Temperature.
MIL-T-83133	- Turbine Fuel Aviation, Kerosene, Type Grade JP-8.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods.
MIL-STD-1474	- Noise Limits for Army Materiel.

HANDBOOKS

MILITARY

MIL-HDBK-113	- Guide for the Selection of Lubricants, Functional Fluids, Preservatives and Specialty Products for Use in Ground Equipment systems.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.1.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

TA13228E1850	- Arctic Fuel System Supply Point.
13222E8219	- Coupling, Rail Tanker, NATO.
13222E8212	- Coupling Set, NATO, Tank Truck Adapter.
13229E2400	- Frame Assembly, Extinguisher, 20 Lb, Halon.
13228E1768	- Gasket, Coupling Half, Quick Disconnect: Cam-Locking Type for Arctic Use.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-FS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 910 - Aviation Gasolines.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The Arctic Fuel System Supply Point, hereinafter referred to as "AFSSP," shall be as specified in TA13228E1850. It shall consist of hoses, valves, manifolds, nozzles, various other components and associated tanks, pumps, and filter-separators in quantities specified herein.

3.1.1 Drawings. The drawings forming a part of this document are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, interchangeability, and operation of the AFSSP. Any data (e.g., shop

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drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Reliability. The specified mean-time-between-failure (MTBF) shall be 195 hours when the AFSSP is operated at rated capacity and tested as specified in 4.5.2.1.

3.4 Maintenance ratio. The AFSSP shall have a total maintenance ratio of not greater than 0.16 (see 6.7.3). A maintenance schedule shall be furnished prior to the start of any testing.

3.5 Material. Materials shall be as specified herein and on the drawings. Material not specified shall be selected by the contractor and shall be subjected to all provisions of this specification.

3.5.1 Material deterioration prevention and control. The AFSSP shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the AFSSP may be exposed.

3.5.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.5.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish, or treatment for use with component and subcomponent, and shall make information available, upon request to the contracting officer or the designated representative. Exterior finishes or treatment shall be approved for camouflage use by the contracting officer or the designated representative (see 3.12).

3.5.2. Recovered materials. For the purpose of this requirement, recovered materials are those materials that have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the AFSSP may be newly fabricated from recovered materials to the maximum extent practicable, provided the AFSSP produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the AFSSP.

3.6 Caps and plugs. Each cam-locking fitting shall be equipped with aluminum dust caps or plugs. The caps and plugs for the fittings shall be held in place by a vibration resistant, cam-locking mechanism. The caps and plugs shall be removable and replaceable without tools and shall provide a positive dust seal when in place.

3.7 Safety and noise.

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3.7.1 Safety. The AFSSP shall present no uncontrolled hazards to the users. All shafts or rotating parts of the pumping unit shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the equipment exposed to contact by personnel or which create a fire hazard shall be fully guarded or insulated. Adequate shielding may be used to attain the same result. External surfaces shall be free of sharp edges and corners except where sharp edges and corners are necessary and are not detrimental to safety. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. Requirement 1 of MIL-STD-454 shall apply.

3.7.2 Noise limits. The noise produced by the AFSSP shall conform to MIL-STD-1474 requirements, with the exception of MIL-STD-1474, 5.2, 5.3, and 5.4, when tested in accordance with 4.5.2.5. The provisions of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.3, except the sign will state "HEARING PROTECTION REQUIRED WITHIN (specify) FEET". The sign shall be readable at the distance indicated on the sign.

3.8 Capacity. The AFSSP when assembled as specified herein for operation, and tested as specified in 4.5.2.6.1, shall deliver not less than 600 gpm at the pump discharge manifold.

3.9 Compatibility. The AFSSP components shall be compatible when connected and shall not leak when tested as specified in 4.5.2.6.2. The AFSSP components shall be suitable for use with hydrocarbon fuels conforming to MIL-T-5624, MIL-G-3056, ASTM D 910, MIL-T-83133, and VV-F-800.

3.10 Continuity. When tested as specified in 4.5.2.6.3, the static electric ground assemblies for the pump, filter-separator, and nozzles shall show a continuous electrical circuit.

3.11 Environmental requirements. The AFSSP shall conform to the following environmental requirements.

3.11.1 Operating temperature. The AFSSP shall perform as specified herein at all ambient temperatures between -60 and +95 °F.

3.11.2 Storage temperature. The AFSSP shall not be damaged by storage at all ambient temperatures between -65 and +145 °F

3.12 Treatment and painting. The portions of the AFSSP normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), top coat color shall be camouflage green 383, conforming to MIL-C-46168.

3.13 Workmanship. All parts and components of the AFSSP shall be cleaned and free from extraneous material. Any necessary rework shall be limited to procedures which will not affect the strength or functioning of parts or components.

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3.14 Cam-locking couplings. Cam-locking couplings shall conform to MIL-C-27487. Cam-locking couplings shall be anodized as specified in MIL-A-8625, type II, class I, thickness no less than 0.0004 inch for castings and no less than 0.0007 inch for wrought, and dyed to match color number 34094 of FED-STD-595. All coupling halves shall have vibration resistant cam-arms. The design shall be so that no pins are required to lock the arms in place. Coupling gaskets shall be in accordance with drawing 13228E1768.

3.15 Components. The sizes, types, and quantities of components shall be as specified in parts list in TA13228E1850.

3.16 Accessory items and bulk materials. Accessory items and bulk materials in the quantities shown on the parts list on TA13228E1850 and table I, shall be supplied with each fuel system module.

TABLE I. Accessory items and bulk materials.

Qty req'd	Nomenclature of description	Identification
1	Wrench, coupling, tank car, 1-3/4 width	5120-00-357-8431
2	Coupling, rail tanker NATO	13222E8219
2	Coupling set, NATO, tank truck adapter	13222E8212
1	Clamping tool, strap band, hose, 3/8 to 3/4 width	GGG-C-413
1	Clamping tool, strap band, hose, 3/8 to 5/8 width	5120-00-359-6587
1	Seal, strapping, CRES, type 201, 3/4 width, 100 EA per box	WW-C-440
1	Seal, strapping, CRES, type 201, 1/2 width, 100 ea per box	WW-C-440
1	Strapping, CRES, type 201, 3/4 width, 100 ft. per roll	QQ-S-766
1	Strapping, CRES, type 201, 1/2 width, 100 ft per roll	QQ-S-766
12	Extinguisher, fire, halon 1211, type F	A-A-1108
4	Frame assembly, extinguisher, 20 lb, Halon	13229E2400

3.17 Lubricants. The procedure for the selection of lubricants shall be in accordance with MIL-STD-838, section 5. Lubricants selected shall be in accordance with MIL-HDBK-113, chapter 2. When the specification of the lubricant selected includes a requirement for a Qualified Products List (QPL), the lubricant supplied shall be from a source that is listed on the applicable QPL.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.6).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.7).
d. Inspection comparison (see 4.6).

4.3 First article inspection.

4.3.1 Sampling. The first article items of each component listed in parts list in TA13228E1850 shall be provided for examination and test.

4.3.2 Examination. The first article components provided in accordance with 4.3.1 shall be examined in accordance with the applicable specification requirements for first article examination and as specified in 4.5.1. Components not covered by a specification shall be examined as specified in 4.5.1 and for compliance with applicable MS part number or drawing referred to in parts list in TA13228E1850. Presence of one or more defects in a component shall be cause for rejection of that component.

4.3.3 Tests. The first article components provided in accordance with 4.3.1 covered by a specification shall be tested as specified in 4.5.2.1 through 4.5.2.5. Rejection criteria shall be in accordance with the requirements of the applicable component specification referenced in parts list in TA13228E1850. Selected samples not covered by a specification shall meet the requirements of the applicable MS part number or drawing referred to in parts list in TA13228E1850.

4.4 Quality conformance inspection.

4.4.1 Sampling. Samples shall be taken in accordance with the applicable specification for each component listed in parts list in TA13228E1850. For components not covered by a specification, samples shall be selected in accordance with MIL-STD-105. Sample size shall be determined by using

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MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined in accordance with the applicable specification requirements and as specified in 4.5.1. Components not covered by a specification shall be examined as specified in 4.5.1. Presence of one or more defects in a component shall be cause for rejection of that component.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1 and 4.5.2.2. Rejection criteria shall be in accordance with the requirements of the applicable specification referenced in parts list in TA13228E1850. Selected samples not covered by a specification shall meet the requirements of the applicable MS part number or drawing referred to in parts list in TA13228E1850.

4.5 Inspection procedure.

4.5.1 Examination. The AFSSP shall be examined as specified herein for the following defects:

101. Materials not as specified.
102. Materials not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration in the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, finishes, or treatment.
105. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the AFSSP.
106. Fitting assemblies not as specified.
107. Treatment and painting not as specified.
108. Paint color not as specified.
109. Workmanship factors not as specified.
110. Cam-locking couplings, caps and plugs, missing or not as specified.
111. Anodizing not specified.
112. Components not as specified.
113. Accessory items and bulk materials not as specified.
114. Lubrication procedure not as specified.

4.5.2 Tests. Test shall be performed without shelter and at the climatic conditions existing at the place of test.

4.5.2.1 Reliability. One AFSSP system shall be tested as specified in 4.5.2 for 650 hours with at least 200 hours accumulating on one pump unit. The lower test value shall be 65 hours and the upper test values shall be 195 hours. Acceptance should be on 5 or fewer failures and reject on 6 or more failures (see 6.7.2). Nonconformance to 3.3 shall constitute failure of this test.

4.5.2.2 Maintenance ratio. The maintenance ratio shall be computed during testing. Nonconformance to 3.4 shall constitute failure of this test.

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4.5.2.3 Component test. Unless otherwise specified (see 6.2), each component shall be tested and accepted in accordance with the requirements of the applicable component specification referenced in parts list in TA13228E1850.

4.5.2.4 Functional test. The samples shall be assembled together to form a continuous hose and manifold system. Assembly and disassembly of the test samples shall be accomplished by hand without the use of tools. The hose and manifold assembly shall be subjected to a pressure of 150 pounds per square inch (psi). While under pressure, the entire system shall be examined for leaks, imperfections in any hose or fitting, or slippage or pullout of the hose fittings. Each valve and nozzle shall be individually activated. Inability to assemble the system by hand, evidence of leakage, imperfections in the hose or fittings, slippage or pullout of the hose fittings, or inability to operate any valve or nozzle shall constitute failure of this test.

4.5.2.5 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments; one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.6 Performance.

4.5.2.6.1 Capacity. The AFSSP shall be assembled for operation as specified in the applicable component technical manual. The AFSSP shall be operated for a period of 48 hours and flow rate shall be determined at the beginning of each hour by means of calibrated flowmeters. Nonconformance to 3.8 shall constitute failure of this test.

4.5.2.6.2 Compatibility. With the AFSSP assembled and operating as specified in 4.5.2.6.1, an inspection of all components and fittings shall be made. Nonconformance to 3.9 shall constitute failure of this test.

4.5.2.6.3 Continuity. With the AFSSP assembled and operating as specified in 4.5.2.6.1, an ohmmeter shall be used to determine electrical continuity of less than 500,000 ohms from the origin of the system to each of the refueling nozzles. Nonconformance to 3.10 shall constitute failure of this test.

4.5.2.7 Safety. The AFSSP shall be evaluated throughout testing as specified in 4.5.2. Nonconformance to 3.7.1 shall constitute failure of this test.

4.5.2.8 Environmental.

4.5.2.8.1 High temperature. The AFSSP shall be tested as specified in MIL-STD-810, method 501.2. The maximum ambient storage temperature shall be 145 °F and the maximum operating temperature shall be 95 °F. The AFSSP shall be operated at rated capacity for a period of 24 hours. Nonconformance to 3.11.1 and 3.11.2 shall constitute failure of this test.

4.5.2.8.2 Low temperature. The AFSSP shall be tested as specified in MIL-STD-810, method 502.2. The minimum ambient storage temperature shall be

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-65 °F. The minimum ambient operating temperature shall be -60 °F. The AFSSP shall be operated at rated capacity for a period of 24 hours. Nonconformance to 3.11.1 and 3.11.2 shall constitute failure of this test.

4.6 Inspection comparison. The Government may select AFSSP at any time during the contract production period and subject these AFSSP to the examination and tests specified in 4.5. The inspection will be performed by the Government, at a site selected by the Government, on AFSSP selected at random from those which have been accepted by the Government and will not include the previously inspected first article AFSSP. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional test shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of any AFSSP to meet any requirement specified herein, during and as a result of the examination and tests specified in 4.5, shall be cause for rejection of the AFSSP and shall be cause for refusal by the Government to continue acceptance of production AFSSP until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on AFSSP produced and accepted under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all AFSSP accepted prior to completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging.

4.7.1 First article pack inspection.

4.7.1.1 Examination. The first article pack shall be examined for the defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.1.2 Test. The first article pack of crated components comprising the fuel system packed for level A shall be subjected to the railroad car test as specified for guided-impact tests in accordance with MIL-STD-1186, appendix A. The car shall strike a string of five empty cars with draft gear extended and the brakes set, at a speed of not less than 10 or more than 11 miles per hour. At the conclusion of the first article pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the components shall constitute failure of the first article pack and shall be cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects.

115. Storage tanks not preserved and packed in accordance with the referenced document for levels A or C.
116. Pumps not preserved and packed in accordance with the referenced document for levels A or C.
117. Filter-separators not preserved and packed in accordance with the referenced document for levels A or C.
118. Hose and nozzles not preserved in accordance with the referenced document for levels A or C.
119. Valves and fittings not preserved in accordance with the referenced document for levels A or C.
120. Fittings and hose assemblies not sealed with caps or plugs as specified for levels A or C.
121. Gaskets not preserved as specified for levels A or C.
122. Gaskets and sealing compound not placed together as specified for levels A or C.
123. Consolidation not as specified for levels A or C.
124. Shipping containers not as specified for levels A or C.
125. Blocking, bracing, and tiedowns not as specified for levels A or C.
126. Packing not as specified for levels A or C.
127. Marking missing, illegible, incorrect, or incomplete for levels A or C.

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination and test within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.9). The first article pack may be accomplished utilizing either the first produced fuel system or a production fuel system. If the first article fuel system is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the first article fuel system and the production fuel system.

5.2 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Storage tanks. Each storage tank shall be preserved in accordance with the level A requirements of MIL-T-53101.

5.2.1.2 Pumps. Each pump shall be preserved in accordance with submethod IIA of MIL-P-116.

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5.2.1.3 Filter-separators. Each filter-separator shall be preserved in accordance with the applicable MIL-F-53091.

5.2.1.4 Hoses and nozzles. Hoses with attached fittings and nozzles shall each be preserved in accordance with the level A preservation requirements of MIL-H-775.

5.2.1.5 Valves and fittings. Valves and fittings shall be preserved in accordance with the level A preservation requirements of MIL-V-3.

5.2.1.6 Caps and plugs. The aluminum caps and plugs shall be placed at random on the fittings and hose assemblies. The remainder of the fittings or hose assemblies shall be sealed with plastic caps or plugs conforming to MIL-C-5501.

5.2.1.7 Gaskets and sealing compound. Gaskets shall be individually preserved, in a flat position, in accordance with MIL-P-116, method IC-1 or IC-3. The gaskets and the sealing compound shall then be placed together in a close-fitting box conforming to PPP-B-636, W6c. The sealing compound shall be secured within the box in a manner to avoid damage to the gaskets.

5.2.1.8 Tools. The preservative application criteria and applicable method(s) of preservation contained in MIL-P-116 shall be used to preserve the tools.

5.2.1.9 Nuts, screws, and washers. Loose nuts, screws, and washers of like size shall be preserved together in accordance with MIL-P-116, method IC-1 or IC-3 and then all placed together in a close-fitting box conforming to PPP-B-636, W6c, style optional.

5.2.1.10 Manifold assembly. The manifold assembly shall be placed in a box conforming to PPP-B-601, style I. Blocking and bracing shall be in accordance with MIL-STD-1186 to prevent movement.

5.2.1.11 Consolidation. Preserved and unpreserved components, except for the tanks, pumps and filter separators, shall be consolidated together in boxes conforming to PPP-B-601, overseas type, style optional. When practical, the boxes shall contain only components of like description. The boxes shall be of a size and configuration to most conveniently accomplish the packing specified in 5.3.1. The components shall be blocked, braced, anchored, or cushioned within the boxes in accordance with MIL-STD-1186 to prevent movement or damage to the contents. Box closure shall be in accordance with the appendix to the applicable box specification. Strapping shall not be required.

5.2.2 Level C.

5.2.2.1 Tanks. Each tank shall be preserved in accordance with level C requirements of MIL-T-53101.

5.2.2.2 Pumps. Each pump shall be filled with the proper operating oil and covered with a waterproof barrier shroud to prevent dirt and water from entering the pump. The shroud shall be secured at the base of the pump.

5.2.2.3 Filter-separators. Each filter-separator shall be preserved in accordance with requirements of MIL-F-53091.

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5.2.2.4 Remaining components. The remaining components shall be preserved as specified for level A.

5.2.2.5 Consolidation. Consolidation shall be as specified for level A, except the box shall be domestic type.

5.3 Packing. Packing shall be level A or C as specified (see 6.2).

5.3.1 Level A. The components of the fuel system, except for the storage tanks, pumps, and filter-separators (see 5.3.1.1 and 5.3.1.2), preserved as specified in 5.2, shall be packed together in a crate conforming to MIL-C-104, type II, class 2, style a. The contents shall be secured to the crate base to provide minimum cube and shall be blocked, braced, and anchored in accordance with the appendix to the applicable crate specification and in accordance with MIL-STD-1186.

5.3.1.1 Storage tanks and filter-separators. The storage tanks and filter-separators shall be packed in accordance with the level A requirements of MIL-T-53101 and MIL-F-53091, respectively. The storage tanks and filter-separators shall be shipped separately from, but at the same time as, the crate containing the other components of the fuel system.

5.3.1.2 Pumps. Each pump shall be packed in a box conforming to PPP-B-601, overseas type, style a, modified as described herein. The containers shall be designed to be reusable by assembling the sides, end, and top as a reusable box cover. Battens, 2 by 2 inches in cross section, shall be positioned on the upper surface of the box base to provide a means to position the cover of the box properly and prevent lateral and longitudinal movement of the cover. Battens shall be located a distance from the edges of the box base equal to the thickness of the cleat and plywood of each side and end panel. The cover shall be secured to the container base with strapping conforming to QQ-S-781, class 1, type I or IV, finish B, size applicable. Strapping shall encircle the box girthwise both laterally and longitudinally; those straps perpendicular to the skids shall be placed between the skids and the lower surface of the box bottom panel.

5.3.2 Level C. The components of the fuel system, except for the tanks, pumps, and filter-separators, preserved as specified in 5.2, shall be packed as specified in level A.

5.3.2.1 Storage tanks and filter-separators. The storage tanks and filter-separators shall be packed in accordance with level C requirements of MIL-T-53101 and MIL-F-53091, respectively. The storage tanks and filter-separators shall be shipped separately from, but simultaneously with the crate containing the other components of the fuel system.

5.3.2.2 Pumps. Each pump shall be packed as specified for level A, except the boxes shall be domestic type.

5.4 Marking. Marking for level A and C shall be in accordance with MIL-STD-129.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The AFSSP is intended for use at low temperature (-60 °F) environment to provide bulk and retail fuel support, fill 5- and 55-gallon containers, and perform the following functions:

a. Receive fuel from tank trucks to fill the storage tanks.
b. Pump fuel from the storage tanks to the dispensing points.
c. Bypass the storage tanks and pump directly from the tank trucks to the dispensing points.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents references (see 2.1.1 and 2.2).
c. When a first article is not required for inspection and approval (see 3.2).
d. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and where, and which examination and tests shall be conducted by the contractor (see 3.2 and 4.3).
e. Color required when other than as specified (see 3.12).
f. When each component is not required to be tested and accepted in accordance with the requirements of the applicable component specification referenced in parts list in TA13228E1850 (see 4.5.2.3).
g. The first article pack required (see 5.1).
h. Level of preservation and packing required (see 5.2 and 5.3).
i. When the 3,000, 10,000 and 20,000 gallon tanks will be furnished by the contractor (see 6.8).
j. When the 600 GPM pumps will be furnished by the contractor (see 6.8).
k. When the 200 GPM filter-separator will be furnished by the contractor (see 6.8).

6.3 First article. When a first article inspection is required, the AFSSP should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.4. Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 Lubricants. MIL-STD-838 prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of nonstandard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-VF, Fort Belvoir, VA 22026-5606, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

6.6 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.7 Definitions. The following definition applies to the text of this specification.

6.7.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of this specification. QAPs indicate the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.7.2 Failure. A failure is defined as any malfunction which causes or may cause:

a. Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
b. Damage to the AFSSP system by continued operation.
c. Serious personnel safety hazards

6.7.3 Maintenance ratio. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before- and after-operation checks are excluded.

6.8 Storage tanks, pumps, and filter-separators. Unless otherwise specified (6.2), the 3,000, 10,000 and 20,000 gallon tanks, 600 GPM pumps, and 200 GPM filter-separators listed in parts list in TA13228E1850 shall not be furnished by the contractor.

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6.9 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the fuel systems in accordance with this specification.

6.10 Subject term (key word) listing.

Aircraft
Berm liner
Filter-separators
Fuel support, bulk
Fueling
Ground equipment
Hoses
Low temperature
Nozzles
Pumps
Refueling
Retail fuel support
Tank collapsible
Valves

Custodian:
Army - ME

Preparing activity:
Army - ME

(Project 4930-A274)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-A-53100(ME)	90/02/07

3. DOCUMENT TITLE
Arctic Fuel System Supply Point (AFSSP)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

[redacted]

6. PREPARING ACTIVITY

a. NAME

b. TELEPHONE (Include Area Code)
(1) Commercial (2) AUTOVON
(703) 664-5717 354-5717

ADDRESS (Include Zip Code)
US Army Belvoir RDE Center
ATTN: STRBE-TSE
Ft. Belvoir, VA 22060-5606

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